

Course Code: BCE-P527**Course Name: MACHINE LEARNING – I**

MM: 100 Time: 3 Hr. L T P 3 0 0	Sessional:30 ESE:70 Credit :3
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Prerequisites:	Understanding of Basic Programming Concept and Mathematics (probability and statistics).
Objectives:	• To learn the fundamentals of Machine Learning. • To understand basic component of an intelligence system. • To explore applications of machine learning. • To understand different types of machine learning algorithms and tools. • To learn how to use machine learning model to solve real world problem.

NOTE:	The question paper shall consist of two sections A and B. Section A contains 10 short type questions of 6 marks each and student shall be required to attempt any five questions. Section B contains 8 long type questions of ten marks each and student shall be required to attempt any four questions. Questions shall be uniformly distributed from the entire syllabus.
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Module	Course Content	No. of Hours	POs mapped	PSOs mapped
<i>Module-1</i>	Introduction to Machine Learning Definition and types of learning: supervised, unsupervised, semi-supervised, reinforcement, Applications of ML (healthcare, finance etc) in real-world scenarios. Tools, Framework, Libraries for ML: Numpy, Pandas, Scikit-learn, matplotlib, jupyter notebook, google colab, Keras, Pytorch	08	PO1/ PO2/ PO3	PSO1/ PSO2
<i>Module-2</i>	Basic concept of Probability theory and Linear Algebra, Bias, Variance, Bias-Variance trade-off, overfitting and underfitting. Pre-processing of data - Data cleaning, wrangling and filtering, Handling missing and categorical data, Data scaling, Feature extraction and selection, covariance matrix, Dimensionality Reduction, Train-Test splitting strategy, Training Set, Validation Set, Test Set, Importance of cross validation – Holdout Method and K-fold cross validation.	08	PO1/ PO2/ PO3	PSO1/ PSO2
<i>Module-3</i>	Introduction to performance metrics for Machine Learning Algorithm – Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Confusion Matrix, Classification Accuracy, Classification Report (Precision, Recall/Sensitivity, Specificity, F1-Score, Area Under ROC curve).	08	PO1/ PO2/ PO3	PSO1/ PSO2
<i>Module-4</i>	Introduction to Linear Regression, cost Function and its intuition, Gradient Descent, simple linear regression, multivariate linear regression, Regularization, Model Evaluation. Real world case-study (House price prediction, Startup Profit predication, student performance prediction etc).	08	PO1/ PO2/ PO3/ PO4	PSO1/ PSO2
<i>Module-5</i>	. Introduction to classification problems and Types of classification - Binary Classification, Multi-Class Classification, Imbalanced Classification, Introduction to Logistic Regression (with Sigmoid Function) Real-world applications case study (Spam Detection, Medical Diagnosis, Fraud Detection, Digit Recognition,	08	PO1/ PO2/ PO3/ PO4	PSO1/ PSO2

	Animal Species Classification)			
Total No. of Hours		40		

Learning Outcomes:	- List various approaches of Machine Learning. - Describe machine learning algorithms to solve the real-world problems. - Develop Hypothesis and machine learning models. - Identify appropriate models for solving machine learning problems. - Apply learning techniques to solve real world machine learning problems.
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Suggested books:

S. No.	Name of Authors /Books /Publisher/Year
1.	Introduction to Machine Learning, Alpaydin, E., MIT Press, 2004.
2.	Introduction to Machine Learning with Python: A Guide for Data Scientists, Andreas C. Müller and Sarah Guido, O'Reilly, 2016.
3.	Elements of Machine Learning, Pat Langley Morgan Kaufmann Publishers, Inc. 1995.
4.	The elements of statistical learning, Friedman, Springer series in statistics, 2001.
5.	The Hundred-page Machine Learning Book, AndriyBurkov, 2019
6.	Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, AurelienGeron, O'Reilly, 2017.
7.	Introduction to Machine Learning, Alpaydin, E., MIT Press, 2004.

CO-PO/PSO MAPPING															
Course Outcomes (COs)	Program Outcomes (POs)												Program Specific Outcomes (PSOs)		
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	
CO1	✓	✓	✓	✓									✓	✓	
CO2	✓	✓	✓	✓									✓	✓	
CO3	✓	✓	✓										✓	✓	
CO4	✓	✓	✓	✓									✓	✓	
CO5	✓	✓	✓										✓	✓	